



Viral mitochondriopathy in COVID-19

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ABSTRACT

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), which causes coronavirus disease 2019 (COVID-19), disrupts cellular mitochondria, leading to widespread chronic inflammation and multi-organ dysfunction. Viral proteins cause mitochondrial bioenergetic collapse, disrupt mitochondrial dynamics, and impair ionic homeostasis, while avoiding antiviral defenses, including mitochondrial antiviral signaling. These changes drive both acute COVID-19 and its longer-term effects, known as “long COVID”. This review examines new findings on the mechanisms by which SARS-CoV-2 affects mitochondria and for the impact on chronic immunity, long-term health risks, and potential treatments.

1. Introduction

The coronavirus disease 2019 (COVID-19) pandemic, caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), had a global impact, with diverse clinical outcomes ranging from asymptomatic infection to severe respiratory failure and death [1]. Severe COVID-19 often causes pneumonia and acute respiratory distress syndrome, which can lead to sepsis, organ failure, or death [2]. The severity varies according to age, existing health conditions, and immune responses. In addition to respiratory involvement, mitochondrial dysfunction may be a central pathological feature of COVID-19 that

contributes to systemic inflammation, organ failure, and long-term sequelae [3].

SARS-CoV-2, a positive-sense single-stranded RNA virus [4], produces three types of proteins: structural (spike glycoproteins, membrane proteins, and envelope proteins), nonstructural (NSPs, 16 in total) [5], and accessory proteins (Fig. 1). These proteins interact with host organelles, particularly the mitochondria and endoplasmic reticulum (ER). In the ER, open reading frame (ORF)3a and ORF8 disrupt normal functions and cause stress, whereas the spike protein undergoes processing [6]. In the mitochondria, ORF9b localizes to the mitochondria-associated membrane (MAM), causing the fragmentation

Abbreviations: AGE, advanced glycation end product; CSF, colony stimulating factor; DAMP, damage-associated molecular pattern; eNOS, endothelial nitric oxide synthase; ER, endoplasmic reticulum; GM-CSF, granulocyte-macrophage colony stimulating factor; IFN, interferon; LDH, lactate dehydrogenase; LPS, lipopolysaccharide; MAM, mitochondria-associated membrane; MAVS, mitochondrial antiviral signaling; MPO, myeloperoxidase; mtDNA, mitochondrial DNA; mtROS, mitochondrial reactive oxygen species; NSP, nonstructural protein; ORF, open reading frame; OXPHOS, oxidative phosphorylation; PDC, pyruvate dehydrogenase complex; RLR, RIG-I-like receptor; ROS, reactive oxygen species; TLR, Toll-like receptor; TNF, tumor necrosis factor.

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and inhibition of immune signaling [7]. ORF3a disrupts calcium homeostasis, and the nucleocapsid protein affect mitochondrial metabolism [8]. These interactions compromise mitochondrial integrity by targeting oxidative phosphorylation (OXPHOS) complexes, inducing calcium dysregulation, and modulating mitochondrial permeability transition pores [9]. Specific viral components, including NSP6, ORF3a, and ORF9b, disrupt mitochondrial architecture and function, promoting viral replication at the expense of host cell viability [9]. Additionally, SARS-CoV-2-induced mitochondrial dysfunction drives systemic metabolic reprogramming and interferes with antiviral immune pathways [10], notably the mitochondrial antiviral signaling (MAVS) pathway [11]. Long-term impairment of mitochondrial function and chronic inflammation, which may persist for weeks or months after the acute phase, are associated with long COVID [12]. This review explores how mitochondrial perturbations contribute to the acute severity and chronic aftermath of COVID-19, including the development of long COVID.

1.1. Disruption of mitochondrial function

Mitochondria regulate cellular metabolism, immune responses, and apoptotic pathways. SARS-CoV-2 infection perturbs these functions by inducing structural and molecular alterations in mitochondrial membranes, disrupting mitochondrial DNA (mtDNA) integrity, and impairing ATP production [13]. In patients with severe COVID-19, circulating mitochondrial proteins, such as cytochromes and ribosomal subunits, are markedly elevated, indicating systemic mitochondrial injury [14].

SARS-CoV-2 infects the alveolar epithelial cells, causing mitochondrial dysfunction and calcium uniporter damage. Calcium signaling controls several key processes, including: viral replication, protein synthesis, metabolism, and cell death. SARS-CoV-2 disrupts calcium homeostasis and affects the infection progresses [15]. The spike protein of the virus binds to angiotensin-converting enzyme 2 (ACE2), which activates Piezo1 and Orai1 channels, leading to increased intracellular calcium levels [16]. However, high calcium levels reduce the fusion activity of the spike proteins [17]. ORF3a in A549 pulmonary epithelial cells increases intracellular calcium levels 18-fold after 48 h [18]. In human embryonic kidney 293 T cells, ORF3a increases mitochondrial calcium via the MCUi11-sensitive uniporter and triggers nucleotide-binding oligomerization domain (NOD)-, leucine-rich repeat

(LRR)-, and pyrin domain-containing protein 3 (NLRP3) inflammasome activation through mitochondrial reactive oxygen species (ROS; mtROS)-induced mtDNA release [19]. ORF3a also promotes interleukin (IL-1 β) secretion through ion disruption and ROS production, activating the NLRP3 inflammasome [20].

1.2. COVID-19 in different mitochondrial groups

SARS-CoV-2 affects mitochondria in a tissue-specific manner, therefore, liver and brain mitochondria respond differently to infection due to their inherent metabolic functions and sensitivity to viral proteins and inflammation. Mitochondria exhibit structural and functional differences in different organs based on tissue-specific metabolic needs [21] (Fig. 2). Liver mitochondria excel in metabolic processes, such as gluconeogenesis and detoxification, requiring flexibility and calcium buffering [22]. Brain mitochondria focus on ATP production for neuronal energy needs and have stricter calcium regulation to prevent excitotoxicity, while being more vulnerable to oxidative stress [23].

Brain mitochondria in COVID-19 are highly sensitive to oxidative stress and calcium imbalance [24]. SARS-CoV-2 proteins impair mitochondrial function by disrupting ER-mitochondrial calcium signaling, increasing ROS production, and altering neuronal mitochondrial dynamics (fission/fusion) [24,25]. These disruptions cause neuroinflammation and neurodegeneration, potentially contributing to “long COVID” cognitive symptoms, and neuronal energy failure, particularly in high-energy demand regions, such as the cortex and hippocampus [25]. Viral infection and systemic inflammation lead to mitochondrial swelling, cristae disruption, impaired β -oxidation, reduced ATP production, and increased susceptibility to lipotoxicity and steatosis [26]. Cytokine storms and hypoxia can further damage hepatocyte mitochondria. The consequences range from mild to severe liver damage and changes in serum transaminases (ALT/AST), which are possibly related to metabolic dysfunction-associated fatty liver disease [27].

1.3. Metabolic alterations

SARS-CoV-2 infection increases ROS levels, alters calcium flux, and modifies mitochondrial morphology [28]. Initially, the virus enhances mitochondrial respiration to support its replication [10]. However,

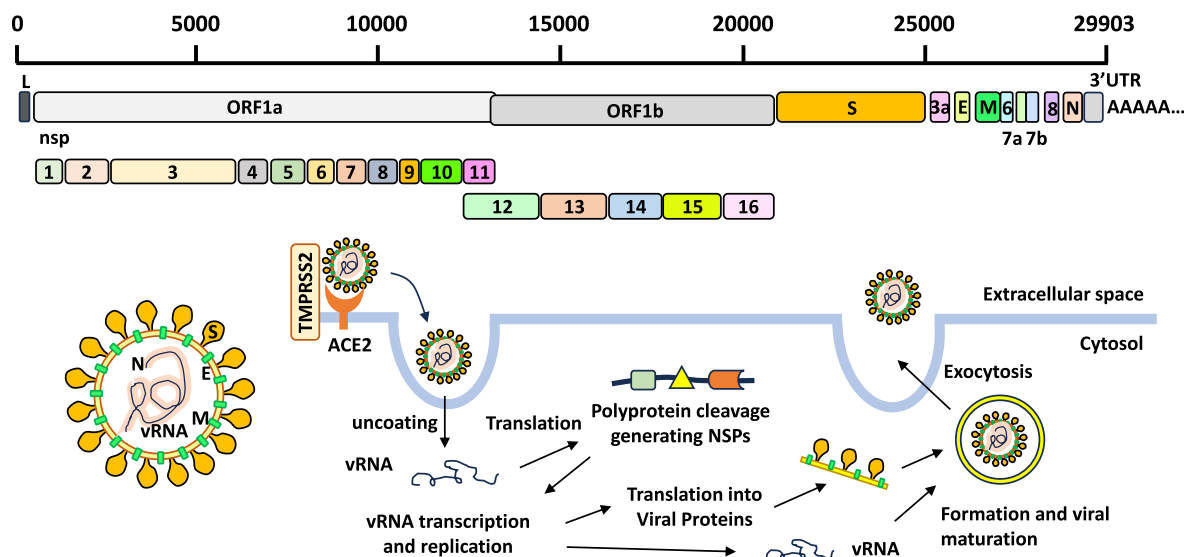


Fig. 1. SARS-CoV-2 genome organization, structural proteins, and replication cycle.

The SARS-CoV-2 genome encodes structural [spike (S), envelope (E), membrane (M), nucleoprotein (N)], nonstructural (NSP1–16), and accessory proteins. The virus enters cells via angiotensin-converting enzyme 2 (ACE2) and transmembrane serine protease 2 (TMPRSS2) to release its RNA genome. This genome then undergoes replication, and virion assembly occurs within the endoplasmic reticulum–Golgi intermediate compartments.

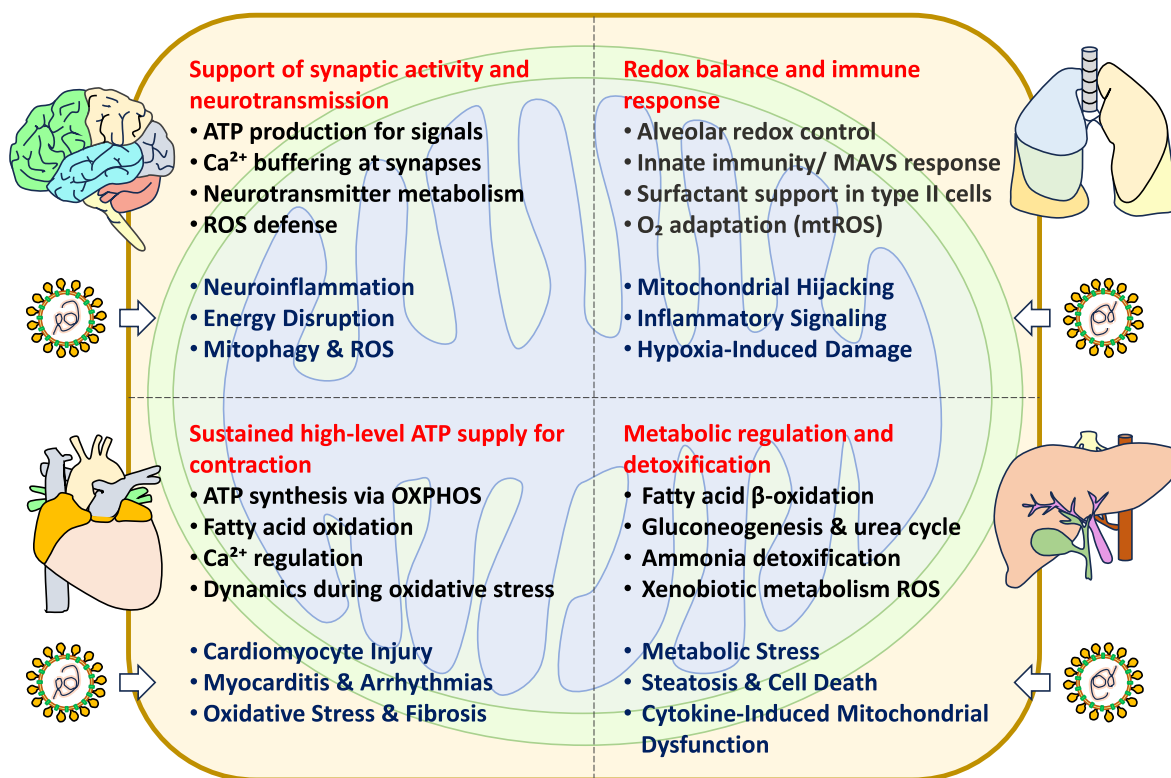


Fig. 2. Mitochondrial special roles, key functions, and SARS-CoV-2-induced mitochondrial disruption across different organs.

Mitochondria in various organs, such as brain, lungs, heart, and liver, manifest structural and functional changes. SARS-CoV-2 can affect these organs by disrupting mitochondrial function, either directly or indirectly through inflammation and oxidative stress.

ATP, adenosine triphosphate; MAVS, mitochondrial antiviral signaling; OXPHOS, oxidative phosphorylation; ROS, reactive oxygen species.

prolonged mitochondrial engagement leads to OXPHOS collapse [29], ATP depletion [30], and oxidative damage [31]. These disruptions impair immune resolution and promote persistent inflammation, contributing to multiorgan involvement and long COVID. SARS-CoV-2 reprograms host metabolism through coordinated disruption of mitochondrial bioenergetics and lipid flux [32] (Fig. 3). SARS-CoV-2 orchestrates widespread metabolic reprogramming that promotes viral replication. One prominent alteration is the shift from OXPHOS to glycolysis (Warburg effect), particularly in endothelial and immune cells [33]. This shift enhances the production of inflammatory mediators and fuels viral biosynthesis [34]. Peripheral immune cells from patients with COVID-19 exhibit elevated glycolytic flux and mitochondrial dysfunction, which are correlated with disease severity [35]. Increased lactate dehydrogenase (LDH) activity, reflecting anaerobic metabolism, is associated with poor prognosis [36]. Notably, lactate accumulation suppresses type I interferon (IFN-I) production, reducing antiviral response.

The virus also perturbs lipid metabolism, inducing lipid droplet biogenesis and co-opting lipid trafficking machinery to support virion assembly. Spike proteins enhance ATP production and mitochondrial respiration by increasing the expression of fatty acid transport regulators and inducing a more negative mitochondrial membrane potential [37]. Additionally, disruptions in amino acid metabolism, particularly in tryptophan pathways [38], alter immune signaling and neurotransmitter synthesis. These cumulative changes result in a metabolic environment that favors viral persistence and immune dysregulation [39]. Inhibiting fatty acid synthase and restoring lipid catabolism through the activation of AMP-activated protein kinase (AMPK) can impede SARS-CoV-2 replication [40].

SARS-CoV-2 proteins alter cellular metabolism by increasing the levels of pyruvate kinase muscle isoform 2. This promotes the accumulation of advanced glycation end-products (AGEs). The abnormally

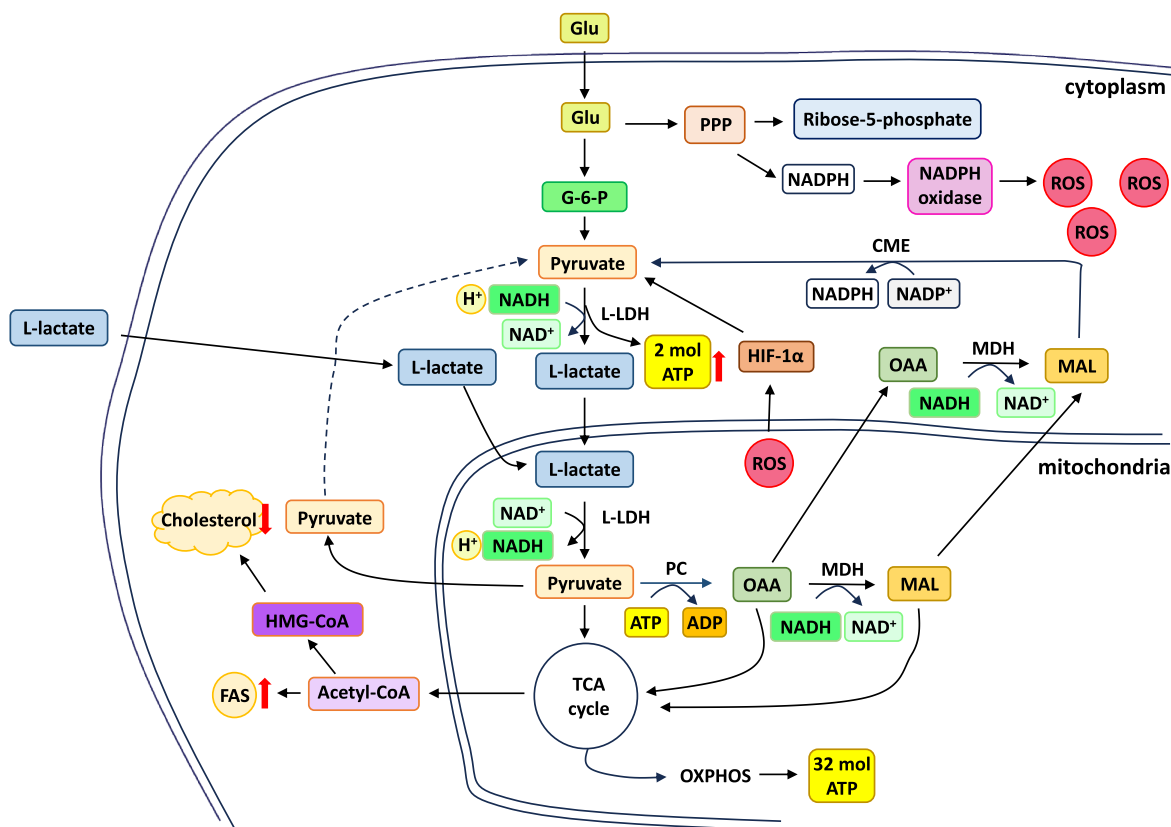
produced AGEs bind to their receptors (RAGEs), activating pro-inflammatory genes, such as *IL-1b* and *IL-6*, which worsen hypoxia and induce aging [41]. In patients with severe COVID-19, $\text{CD14}^+\text{CD16}^-$ monocytes activate the NLRP3 inflammasome. This leads to increased $\text{IL-1}\beta$ production through caspase-1/apoptosis-associated speck-like protein containing a caspase recruitment domain (ASC) speck formation, along with abnormal mitochondrial superoxide production, and lipid peroxidation [42].

1.4. Abnormalities mitochondrial dynamics

Mitochondrial dynamics regulate the mitochondrial shape and function through fusion, fission, biogenesis, and mitophagy. SARS-CoV-2 infection disrupts mitochondrial function by causing membrane depolarization and increasing ROS release [43]. This dysfunction leads to organ failure, inflammation, and mortality in patients with COVID-19 and may explain the persistent fatigue observed in long COVID [12, 44]. SARS-CoV-2 disrupts energy production by increasing the expression of mitochondrial dynamics proteins: PINK1, Parkin, and MFN2 [10, 43, 45]. Patients with lung complications exhibited elevated PINK1, DNML1, and MFN2 levels [46]. ORF3a triggers mitochondrial fission and cell death [47], while ORF9b promotes mitochondrial fusion and cell survival [48].

1.5. Programmed cell death and release of damage-associated molecular patterns

SARS-CoV-2 infection activates mTORC1, which triggers the IRE1/JNK pathway and regulates apoptosis in distinct ways during the early and persistent infection stages [48, 49]. Viral proteins influence cell death through several mechanisms: they increase apoptosis-inducing factor expression, activate caspase 7 [10], and alter mitochondrial



Acetyl-CoA, Acetyl-coenzyme A; ADP, adenosine diphosphate; ATP, adenosine triphosphate; FAS, fatty acid synthase; G-6-P, Glucose 6-phosphate; Glu, glucose; HIF-1 α , Hypoxia-inducible factor 1-alpha; HMG-CoA, 3-hydroxy-3-methylglutaryl-coenzyme A; L-LDH, L-lactate dehydrogenase; MAL, Malic acid; MDH, malate dehydrogenase; NAD⁺, Nicotinamide adenine dinucleotide (oxidized form); NADH, Nicotinamide adenine dinucleotide (reduced form); NADPH, nicotinamide adenine dinucleotide phosphate (reduced form); NADP⁺, nicotinamide adenine dinucleotide phosphate (oxidized form); OAA, Oxaloacetic acid; PC, phosphocreatine; PPP, pentose phosphate pathway; ROS, reactive oxygen species; TCA, tricarboxylic acid.

Damaged mitochondria increase ROS and pro-inflammatory cytokines, releasing damage-associated molecular patterns (DAMPs) into the cytoplasm, which triggers cell death and inflammation. This process leads to widespread damage, including oxidative stress, hyperferritinemia, and thrombosis. When released by damaged cells, DAMPs bind to pattern recognition receptors on immune cells [55], activating inflammatory pathways through Toll-like (TLRs), RIG-I-like, and NOD-like receptors. This binding activates NF- κ B and inflammasome pathways, which release pro-inflammatory cytokines [55]. SARS-CoV-2 infection causes cell death, releasing DAMPs, such as HMGB1 and heat shock proteins [56], while viral RNA and host DNA serve as additional inflammatory triggers. NSP4 and ORF9b damage the mitochondria by forming membrane pores and releasing mtDNA-containing vesicles [57]. NSP4 creates these pores through interactions with BAK, whereas ORF9b inhibits MCL-1 (an anti-apoptotic member of the BCL2 protein family), leading to mtDNA release [57]. ORF9b supports viral

1.6. MAVS pathway interference

1.7. MAVS pathway

Tumor necrosis factor (TNF) receptor-associated factor 3 (TRAF3)-interacting protein 3 (TRAF3IP3) accumulates in the mitochondria.

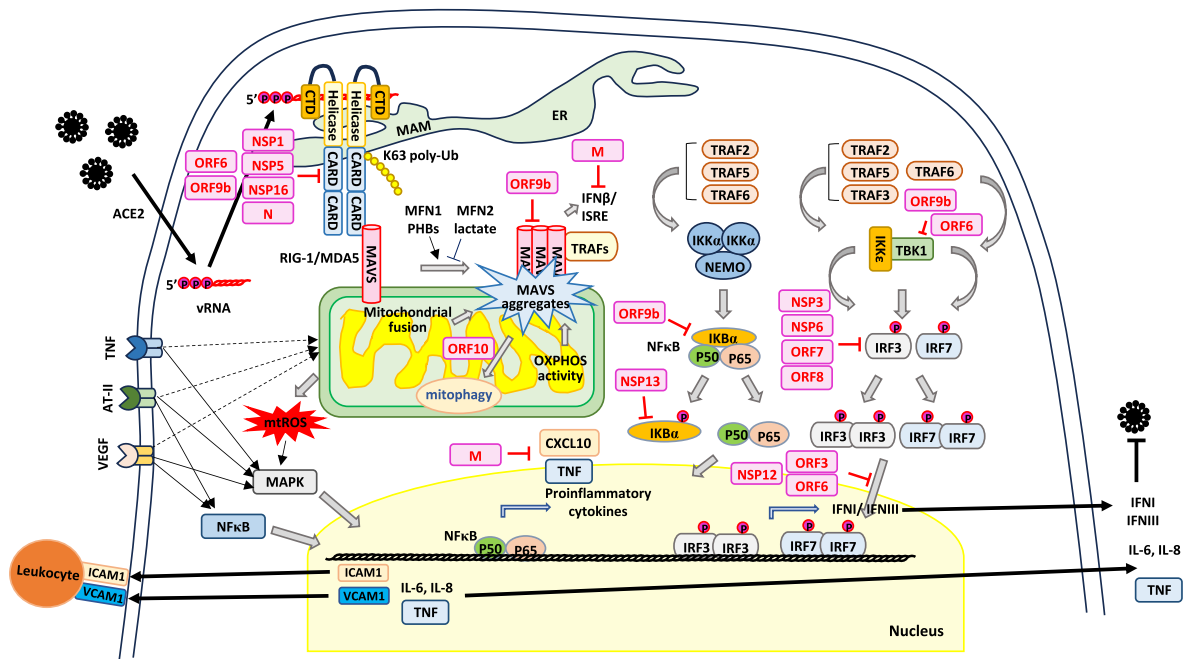


Fig. 4. Mitochondrial antiviral signaling and viral evasion mechanisms.

Mitochondrial antiviral signaling (MAVS) provides an antiviral defence system with the assistance of RIG-I, MDA5, and LGP2 receptors. When the viral genome triggers conformational changes in RIG-I and MDA5 in the outer mitochondrial membrane, these receptors use their CARD domains to interact with MAVS. This interaction prompts the MAVS proteins to aggregate into MAVS signalosomes, which activates TANK-binding kinase (TBK1) and IRFs. TRAF3 and TRAF6 contribute to antiviral protection by activating NF-κB and IRF. Nuclear translocation of NF-κB initiates pro-inflammatory cytokine gene expression, whereas IRF translocation enhances interferon production. SARS-CoV-2 counteracts this defense through several mechanisms. Its ORF-9b protein inhibits RIG-I, MDA5, MAVS, and TBK1, thereby blocking IRF signaling and interferon activation. Viral membrane proteins downregulate MAVS-related pathways and restrict TRAF3, TBK1, and IRF3 recruitment to the MAVS complex. Additionally, ORF-6 blocks interferon induction by MDA5, MAVS, and TBK1.

ACE2, Angiotensin-converting enzyme 2; AT-II, Angiotensin type 2 receptor; CARD, Caspase-recruitment domain; CTD, Carboxy-terminal domain; CXCL10, C-X-C motif chemokine ligand 10; ER, Endoplasmic reticulum; ICAM1, Intercellular adhesion molecule 1; IFN, Interferon; IKKα, Nuclear factor of kappa light polypeptide gene enhancer in B-cells inhibitor alpha; IKKβ, Inhibitor of NF-κB Kinase α; IL, Interleukin; IRF, Interferon regulatory factor; ISRE, Interferon-stimulated response element; K63 poly-Ub, K63-linked ubiquitination; LGP2, Laboratory of genetics and physiology 2; M, Membrane protein; MAM, Mitochondria-associated membranes; MDA5, Melanoma differentiation-associated protein 5; MFN, Mitofusin; N, Nucleocapsid protein; NEMO, NF-κB essential modulator; NF-κB, Nuclear factor kappa B; NSP, Non-structural proteins; PHB, Prohibitin; RIG-I, Retinoic acid-inducible gene I; TBK1, TANK-binding kinase 1; TNF, Tumor necrosis factor; ORF, Open reading frame; TRAF, TNF receptor-associated factor; VCAM1, Vascular cell adhesion protein 1; VEGF, Vascular endothelial growth factor; vRNA, Viral RNA.

promotes TRAF3 recruitment, and facilitates MAVS recruitment, which activates TBK1-IRF3 signaling. TRAF3IP3 serves as a crucial link in RIG-I-MAVS signaling and enhances the antiviral response. In MAM, MAVS interacts with glutamine-fructose-6-phosphate aminotransferase, the first rate-limiting enzyme in the hexosamine biosynthetic pathway. MAVS signaling forms at the MAM, recruits TRAF6 and TRAF2, and links RLR signaling to glucose metabolism [63]. Specific regions of MAVS activate the transcription factors, IRF3 and NF-κB, which recruit TRAF for downstream signaling. These factors trigger the production of IFN-I (including IFN-α and IFN-β), IFN-III (including IFN-λ), and other antiviral cytokines. This process establishes an antiviral state, strengthens the immune response against infections, and prevents excessive inflammation [64]. Finally, MAVS can directly interact with LC3 through its LC3 binding motif, thereby maintaining mitochondrial homeostasis through mitophagy.

1.8. SARS-CoV-2 disrupts MAV

SARS-CoV-2 modifies the mitochondrial dynamics and targets MAVS to evade host immunity and facilitate replication [65]. Mitochondria-localized ORF9b enhances the interaction between poly (rC)-binding protein 2 (PCBP2) and HECT domain E3 ligase (AIP4), thereby inhibiting the MAVS signaling pathway. SARS-CoV-2 membrane protein prevents MAVS protein aggregation, whereas ORF-6 disrupts MAVS activation [61]. Additionally, the ORF3c binds to MAVS and phosphoglycerate mutase 5 (PGAM5), triggering MAVS cleavage via

caspase-3 [66]. These combined effects promoted viral replication and impaired mitochondrial function [67].

SARS-CoV-2 infection induces elevated inflammatory cytokine levels and, paradoxically, results in very low levels of IFN-I and IFN-III [68]. Several viral proteins, including ORF3, ORF3b, ORF6, ORF7a, ORF7b, ORF8, and ORF9b, inhibit IFN responses to varying degrees [69]. ORF3c, which localizes to mitochondria, suppresses innate immunity by limiting IFN-β production without affecting NF-κB activation or Janus kinase (JAK)-STAT signaling. ORF7a disrupts IFN-I responses, manipulates the host ubiquitin system to enhance and counteract IFN-I responses [70], reduces antigen presentation, and triggers pro-inflammatory cytokine responses [71]. ORF8 binds to MHC-I, activates the IL-17 pathway, and promotes pro-inflammatory factors [72, 73]. ORF10 binds to NIX, mediates mitophagy-driven degradation of MAVS, and inhibits the IFN-I signaling pathway [53]. Both ORF7a and ORF9b, located in the mitochondria, inhibit RIG-I-MAVS-dependent signaling [12,74]. ORF9b further disrupts MAVS and interferon signaling by promoting the degradation of MAVS, TRAF3, and TRAF6 [75]. SARS-CoV-2 disrupts MAVS signaling through several mechanisms. ORF9b causes MAVS breakdown via PCBP2-AIP4, whereas ORF3c and ORF10 alter its localization and affect mitophagy. These mechanisms suppress IFN-I responses and help viruses evade detection. The viral proteins ORF6, ORF7a, and ORF8, evade immune responses by blocking interferon production and antigen presentation.

1.9. ORF9b disrupts HSP90–TOM70 interaction: impairment of mitochondrial antiviral response and metabolism

ORF9b plays multiple roles in viral replication and immune system evasion [8]. This disrupts MAVS, inhibits IFN production, and promotes mitochondrial fusion, thereby preventing apoptosis. Translocase of the outer mitochondrial membrane 70 (TOM70) is essential for mitochondrial energy metabolism and its dysfunction can lead to lactic acidosis. ORF9b targets TOM70 specifically to suppress IFN-I response [75]. This interaction is regulated by ORF9b phosphorylation at serine residue 53, which interacts with the TOM70 glutamate at residue 477 in infected cells [76]. By binding to the C-terminus of TOM70 (residues 235–608), ORF9b impairs MAVS function [6,75], and triggers both IFN-I inhibition and lactate production. HSP90 interacts with TOM70 and is crucial for TOM70-mediated IFN-I activation. ORF9b competes with HSP90 for TOM70 binding. When ORF9b occupies the C-terminal domain of TOM70, it reduces the Glu-Glu-Val-Asp motif-binding affinity of HSP90 to the N-terminal domain of TOM70 by approximately 29-fold [77]. ORF9b blocks the interaction between HSP90 and TOM70, a key component of IFN signaling and mitochondrial balance, through the MAVS pathway. This disruption weakens early immune responses, which may explain the prolonged viral presence and inflammation observed in long COVID.

1.10. Mitochondrial damage induces inflammation

SARS-CoV-2 induces mitochondrial dysfunction and oxidative stress [78]. In severe cases, elevated mROS levels, and NADPH oxidase activity impair immunity and cause inflammation [79]. The virus increases ROS levels via inflammatory pathways, leading to cytokine storms [80]. Inflammatory cytokines, such as TNF- α and IL-6 boost mROS [81], while increasing viral loads decrease CD14⁺ monocytes [82]. The virus impairs antioxidant responses via hypoxia-inducible factor (HIF)-1 α [83]. The initial infection causes cell death and endothelial damage, triggering inflammation that can lead to organ failure. Severe cases show neutrophil activation and excess pro-inflammatory cytokines that disrupt mitochondrial function [84].

1.11. Persistent inflammatory response

Residual viral particles or RNA can trigger persistent immune responses or symptoms, leading to long-term adverse effects. Mitochondrial dysfunction may cause chronic inflammation, potentially spreading to multiple organs, thus increasing COVID-19 severity. In patients with severe COVID-19, HIF-1 α regulates glycogen phosphorylase L in neutrophils, enhancing glycolysis and glycogen accumulation. This results in higher concentrations of plasma neutrophil extracellular trapmarkers, including myeloperoxidase (MPO), elastase, and MPO-DNA complexes [85]. Key neutrophil function regulators, such as CCL2, CXCL10, CCL20, IL-18, IL-3, IL-6, granulocyte colony-stimulating factor (CSF), granulocyte-macrophage CSF (GM-CSF), TNF- α , and IFN- γ , are significantly increased in the plasma of patients with severe COVID-19 [46,85]. A moderate correlation exists between RAGE and TNF- α levels, as well as between DNM1L and IFN- α levels [46]. These alterations promote inflammatory response. Moreover, GM-CSF and prostaglandin E2 levels are elevated, while IFN- α is decreased in these patients [86]. SARS-CoV-2 activates TLRs and the NLRP3 inflammasome, triggering the production of various inflammatory molecules, including IL-6, IL-1 β , IL-18, IL-8, IL-10, TNF- α , and LDH detoxification [83,87]. This activation leads to significant cellular changes, such as membrane rupture, cytoplasmic swelling, and chromatin disorganization [87]. Nicotine exacerbates these effects by increasing the inflammatory cytokines levels, causing severe cell damage [87]. In SARS-CoV-2-infected endothelial cells, the expression of ACE2 and TMPRSS2 results in mitochondrial dysfunction, TLR9 activation, reduced endothelial nitric oxide synthase (eNOS) activity, and

inhibition of calcium responses. Notably, blocking TLR9 signaling further reduces IL-6 release and prevented eNOS decline, contributing to the severity of COVID-19 symptoms [88].

1.12. Additional factors influencing the inflammatory response

COVID-19 mortality rates are higher in men than in women [89]. Women exhibit stronger immune responses [90], because of differences in sex chromosomes and hormones, such as estrogen, progesterone, and testosterone [91]. Additionally, estrogen enhances antioxidant enzymes, such as manganese superoxide dismutase and GSH-peroxidase, which play crucial roles in mitigating ROS-induced damage [92]. Mitochondria also play a role in sex-based immune variations, which are crucial for effective immune responses [93]. Severe respiratory infections are the primary cause of death, particularly among elderly patients. This increased mortality largely stems from the hypoxic and hyper-inflammatory states linked to COVID-19-related sepsis [94].

Melatonin enhances both humoral and cell-mediated immune responses by promoting the production of inflammatory cytokines and Type 1 T helper (Th1) cells. It shows antiviral activity by inhibiting inflammatory mediators, such as IL-6, IL-1 β , and TNF- α , which are released during severe COVID-19 lung injury [95]. By inducing the expression of the circadian gene *Bmal1* and inhibiting the pyruvate dehydrogenase complex (PDC), melatonin counteracts the viral inhibition of *Bmal1*/PDC. PDC helps convert pyruvate to acetyl-CoA in the mitochondria, supporting the tricarboxylic acid cycle, OXPHOS, and ATP production. These alterations increase intestinal permeability and dysbiosis, suppress short-chain fatty acids, such as butyrate, and increase circulating lipopolysaccharide (LPS) levels. Pineal-produced melatonin inhibits and mitigates these effects, preventing a “reset” of the circadian rhythm in mitochondrial metabolism [96]. Disruptions in butyrate and LPS levels may boost viral replication and worsen host symptoms by interfering with the melatonin pathway [96].

1.13. Limitations and future directions

SARS-CoV-2 proteins (such as ORF9b, NSP4, and membrane proteins) localize to mitochondria and disrupt their function [57,61]; however, the mechanism by which these changes cause chronic inflammation is unclear. The current knowledge mainly comes from laboratory studies and samples from acute cases. We lack long-term patient data showing how mitochondrial problems during infection relate to ongoing inflammation, particularly in long COVID [12,44]. Additionally, standard research models, such as lab-grown cells or mice, may not accurately represent human mitochondria and immune systems. Further research is required to determine mitochondrial responses in the brain, heart, and lungs.

mtDNA damage and disease severity are associated [13], though a clear cause-and-effect relationship in humans remains to be established. Mitochondrial responses vary based on individual factors, such as age, sex, and underlying health conditions. The influence of pre-existing conditions, such as metabolic syndrome or mitochondrial diseases, on viral infection and inflammation remains unclear. The interplay between mitochondria and other cellular processes (including autophagy, ER stress, and inflammasome activation, such as NLRP3) during in SARS-CoV-2 infection also remains unclear [20]. Additionally, whether mitochondrial changes continue after the initial infect, such as in long COVID, is unknown. Future studies should clarify the causal link between mitochondrial injury and long COVID, validate targeted interventions in clinical settings, and explore individual variations in mitochondrial responses to infection. A deeper understanding of these pathways may reveal precise therapies for both COVID-19 and other mitochondria-related diseases.

Targeting mitochondrial pathways offers a promising avenue for reducing excessive inflammation by improving mitochondrial balance. Several strategies may help restore metabolic equilibrium and reduce

long-term complications: mitochondria-targeting antioxidants [such as mitoquinone (MitoQ) and visomitin (SkQ1)] [97,98], compounds that trigger mitophagy (such as PINK1-Parkin pathway activators), metabolic regulators (such as AMPK activators and PPAR agonists) [99], and MAVS pathway stabilizers. Although these mitochondria-focused treatments are promising, more evidence is required to confirm their safety and effectiveness during infection.

2. Conclusion

Mitochondrial dysfunction plays a key role in SARS-CoV-2 pathogenesis, connecting the fields of virology, immunology, and metabolism. The virus hijacks host mitochondria, using them to boost replication, while disrupting immune responses and causing lasting cellular damage. This dysfunction contributes to the development of severe, acute COVID-19 symptoms and long-term complications. Several promising treatments, such as antioxidants, mitophagy modulators, and MAVS stabilizers, target mitochondrial pathways. However, further research is needed to confirm the effectiveness of these treatments and understand how mitochondrial damage affects post-COVID conditions.

CRediT authorship contribution statement

Tsung-Hsien Chen: Writing – review & editing, Writing – original draft, Supervision, Project administration, Formal analysis. **Tien-Hsin Jeng:** Writing – review & editing, Writing – original draft, Formal analysis. **Ming-Yang Lee:** Writing – review & editing. **Hsiang-Chen Wang:** Writing – review & editing. **Kun-Feng Tsai:** Writing – review & editing, Validation, Supervision, Project administration. **Chu-Kuang Chou:** Writing – review & editing, Validation, Supervision, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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